Extra Bite #5: RF Parts Are Not Ideal

Skin effect, parasitic reactance, and "wattless" reactive power. Why a resistor stops being a resistor above 30 MHz.

Group E5D · 12 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

At DC and audio, a resistor is a resistor and a capacitor is a capacitor. At RF those same parts grow extra personalities: resistors look slightly inductive, capacitors hit a self-resonance and start acting like inductors, and inductors hit a self-resonance and start acting like capacitors. E5D is the group that tests whether you understand **why parts misbehave at RF**.

Skin Effect

At DC, current uses the whole cross-section of a wire. At RF, magnetic forces inside the wire push current toward the **surface** – the "skin" of the conductor. With less effective copper carrying the current, **resistance increases as frequency increases**. (E5D01)

Consequence: at RF you pay more loss in a thin solid conductor than in a tube or a braided/silver-plated surface. Skin effect is also the **primary cause of loss in film capacitors at RF**, because the electrode foil sees the same effect. (E5D08)

Real Power vs Reactive Power

Type of power	What it does	Current-vs-voltage phase
Real power	Dissipated as **heat** in resistors. The useful, paid-for work.	In phase (0°)
Reactive power	Stored in magnetic or electric fields and given back. **Wattless, nonproductive.** (E5D12)	**90° out of phase** (E5D03)

In ideal inductors and capacitors, reactive power swings back and forth between the field and the source but is **not dissipated**. (E5D09)

Worked example: real power in a partly-reactive circuit

Circuit: 100-ohm resistor in series with 100-ohm inductive reactance, drawing 1 A.
Real power dissipates only in the resistor: $P = I^2 \times R = 1^2 \times 100 =$ **100 watts**. (E5D11)
The inductor stores reactive power but does not dissipate any of it. So the 100-ohm inductive reactance contributes zero real power.

Parasitic Effects In Real Components

Component	The parasitic that ruins it	Q
Electrolytic capacitor	**Inductance** (long internal foil rolled into a coil). Unsuitable for RF bypass – use ceramic or film for HF and ceramic/multilayer for VHF+.	E5D05
Inductor (any coil)	**Inter-turn capacitance** between adjacent turns creates a parallel-resonant tank with the inductance, giving the coil its **self-resonance frequency**.	E5D06
Any real component	Self-resonance comes from **nominal reactance plus parasitic reactance** combining: a "capacitor" has parasitic inductance, an "inductor" has parasitic capacitance, and they meet.	E5D07

Lead Length At VHF And Microwave

Every inch of component lead is a small inductor. At HF this is negligible; at VHF and above it is dominant.

- **VHF and above:** keep leads short to **minimize inductive reactance**. (E5D02)
- **Microwave:** keep connections short to **reduce phase shift** along the connection. (E5D04)

Conductor Diameter And Electrical Length

The "electrical length" of a wire is how long it acts at the operating frequency – affected by end effects, dielectric, and conductor diameter.

As conductor diameter increases, electrical length increases. (E5D10) That is why fat antenna elements end up slightly shorter in physical length than thin-wire dipole formulas suggest – they are electrically longer per foot, so you need fewer feet to reach the same resonant electrical length.

Mental Cheat Sheet

- Skin effect: RF current flows on the surface; resistance **increases** with frequency.
- Real power = dissipated in R (in phase). Reactive power = stored in L/C (90° out of phase, **wattless**).
- Ideal L and C do not dissipate reactive power; they store and return it.

Common Traps

- Skin effect **increases** resistance with frequency – do not pick the "decreases" answer.
- Reactive power is wattless – the test offers "consumed in inductors and capacitors" as a trap.
- The reactive part of an impedance contributes **zero** real power. $P = I^2R$ uses only the resistance.

- P (real) = I^2R. The reactive part contributes zero to real power.
- Electrolytic caps fail at RF due to **inductance**.
- Film caps lose at RF due to **skin effect**.
- Inductor self-resonance: **inter-turn capacitance** resonating with the nominal L.
- Self-resonance in general = nominal reactance + parasitic reactance.
- VHF+: short leads = less inductive reactance.
- Microwave: short leads = less phase shift.
- Fatter conductor ⇒ longer electrical length.

- Electrolytic cap failure at RF is from **inductance**, not from leakage or dielectric loss.
- Inductor self-resonance is from **inter-turn capacitance**, not "shunt capacitance to ground" or "skin effect."
- Fat conductor ⇒ **longer** electrical length, not shorter or unchanged.

Takeaway: RF parts have a second personality – skin effect raises resistance with frequency, every component has a self-resonance where its parasitic reactance meets its nominal reactance, leads turn into inductors at VHF, and reactive power swings back and forth without ever paying the electric bill.

Bite 5 Question Bank (E5D)

12 questions. Read, decide, check the answer key at the end.

E5D01. What is the result of conductor skin effect?
A. Resistance increases as frequency increases because RF current flows closer to the surface.
B. Resistance decreases as frequency increases because electron mobility increases.
C. Resistance increases as temperature increases because of the change in thermal coefficient.
D. Resistance decreases as temperature increases because of the change in thermal coefficient.

E5D02. Why is it important to keep lead lengths short for components used in circuits for VHF and above?
A. To increase the thermal time constant.
B. To minimize inductive reactance.
C. To maintain component lifetime.
D. All these choices are correct.

E5D03. What is the phase relationship between current and voltage for reactive power?
A. They are out of phase.
B. They are in phase.
C. They are 90 degrees out of phase.
D. They are 45 degrees out of phase.

E5D04. Why are short connections used at microwave frequencies?
A. To increase neutralizing resistance.
B. To reduce phase shift along the connection.
C. To increase compensating capacitance.
D. To reduce noise figure.

E5D05. What parasitic characteristic causes electrolytic capacitors to be unsuitable for use at RF?
A. Skin effect.
B. Shunt capacitance.
C. Inductance.
D. Dielectric leakage.

E5D06. What parasitic characteristic creates an inductor's self-resonance?
A. Skin effect.
B. Dielectric loss.
C. Coupling.
D. Inter-turn capacitance.

E5D07. What combines to create the self-resonance of a component?
A. The component's resistance and reactance.
B. The component's nominal and parasitic reactance.
C. The component's inductance and capacitance.
D. The component's electrical length and impedance.

E5D08. What is the primary cause of loss in film capacitors at RF?
A. Inductance.
B. Dielectric loss.
C. Self-discharge.
D. Skin effect.

E5D09. What happens to reactive power in ideal inductors and capacitors?
A. It is dissipated as heat in the circuit.
B. Energy is stored in magnetic or electric fields, but power is not dissipated.
C. It is canceled by Coulomb forces in the capacitor and inductor.
D. It is dissipated in the formation of inductive and capacitive fields.

E5D10. As a conductor's diameter increases, what is the effect on its electrical length?
A. Thickness has no effect on electrical length.
B. It varies randomly.
C. It decreases.
D. It increases.

E5D11. How much real power is consumed in a circuit consisting of a 100-ohm resistor in series with a 100-ohm inductive reactance drawing 1 ampere?
A. 70.7 watts.
B. 100 watts.
C. 141.4 watts.
D. 200 watts.

E5D12. What is reactive power?
A. Power consumed in circuit Q.
B. Power consumed by an inductor's wire resistance.
C. The power consumed in inductors and capacitors.
D. Wattless, nonproductive power.

Answer Key

E5D01: A	E5D02: B	E5D03: C
E5D04: B	E5D05: C	E5D06: D
E5D07: B	E5D08: D	E5D09: B
E5D10: D	E5D11: B	E5D12: D